

■ Inner

`Inner[f, list1, list2, g]` is a generalization of `Dot` in which f plays the role of multiplication and g of addition.

Example: `Inner[f, {a,b}, {x,y}, g]  $\longrightarrow$  g[f[a, x], f[b, y]]`.

■ `Inner[f, {{a,b},{c,d}}, {x,y}, g]  $\longrightarrow$  {g[f[a, x], f[b, y]], g[f[c, x], f[d, y]]}`. ■ Like `Dot`, `Inner` effectively contracts the last index of the first tensor with the first index of the second tensor. Applying `Inner` to a rank r tensor and a rank s tensor gives a rank $r + s - 1$ tensor. ■ `Inner[f, list1, list2]` uses `Plus` for g .

■ `Inner[f, list1, list2, g, n]` contracts index n of the first tensor with the first index of the second tensor. ■ The heads of `list1` and `list2` must be the same, but need not necessarily be `List`. ■ See page 457. ■ See also: `Outer`, `Thread`.